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Machine Learning & Communication Skills – Project Proposal

Early Collision Risk Forecasting from ESA Conjunction Data Messages

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Text limit:	2 pages; figures excluded from the page count

1 Problem, Motivation, and Research Question

Low Earth Orbit is increasingly congested. When two objects may approach closely, operators receive Conjunction Data Messages (CDMs) with geometry, covariance, and a self-computed collision risk. Over about one week before time of closest approach (TCA), several CDMs arrive; many events that look concerning early become clearly safe near TCA. Operators often begin avoidance planning about two days before TCA [1, 3].

How this project differs from the ESA Kelvins competition. The 2019 Kelvins Collision Avoidance Challenge asked teams to minimise a single leaderboard loss $L = \text{MSE}_{\text{HR}}/F_2$ on early CDMs [1, 3]. We reuse that public dataset and the same early-input constraint (time-to-TCA ≥ 2 days), but our scientific goal is *not* to climb that leaderboard or to reproduce winning competition pipelines. We treat Kelvins as a shared benchmark and prior art: we ask which modelling strategies improve upon a strong last-risk-prediction (LRP) baseline under rare-event metrics, with calibrated high-risk probabilities, explicit analysis of "jump" failures, and a transparent failure atlas. Success is decision-support insight and honest characterisation of uncertainty—not a competition ranking.

Intended impact / outcome. If successful, the project delivers (i) a reproducible event-level protocol comparing course and ambitious models against LRP under F_2 and MSE_{HR}; (ii) evidence on whether a two-stage design—detect high risk, then estimate risk magnitude—better characterises uncertain near-TCA jumps than point estimates alone; and (iii) an actionable failure atlas operators or analysts could use to know *when not to trust* an early forecast. We do not claim operational deployment or autonomous commanding.

Primary RQ. Given CDMs with time-to-TCA ≥ 2 days, which modelling approach best improves upon LRP for final collision risk under rare-event metrics F_2 and MSE_{HR}?

Secondary RQ. Can a calibrated high-risk probability in a two-stage design better characterise uncertain jump events—where risk rises sharply near TCA—than point estimates alone?

Scope and ambition. Academic study on historical anonymised ESA CDMs. We will apply at least one syllabus method and are prepared to experiment with carefully chosen out-of-syllabus tools (e.g. sequence models, modern calibration, selective prediction) when they serve the RQs—without copying Kelvins winning stacks as the claim [3].

2 Data and Exploratory Analysis

We use the ESA Collision Avoidance Challenge dataset on Zenodo (DOI `10.5281/zenodo.4463683`; CC BY 4.0) [2, 3]. The unit is a close-approach *event* (`event_id`): a short CDM series. Each CDM has 103 attributes (geometry, orbit/OD quality, covariances, space weather). The target is final event risk

($\log_{10}$ in the release). High-risk (HR): final $r \geq -6$ (10^{-6} linear) [3].

Sizes. Train: 162,634 CDMs / 13,154 events. Official test: 24,484 early CDMs / 2,167 events with private final labels; 150 HR ($\approx 6.9\%$). Public test features contain only CDMs with time-to-TCA ≥ 2 days (no leakage of near-TCA messages).

EDA findings. Figure 1 shows extreme imbalance (mass of very low finals), so accuracy/overall MSE would mislead. Figure 2 states the truncation protocol. Known limitations [3]: non-i.i.d. train/test construction; mission mix among HR events differs across splits; manoeuvre cases removed/truncated; labels are operational risk estimates, not observed collisions. As a first course-method check (PCA), we inspected low-dimensional structure of numeric CDM features; mission structure and low-risk dominance reinforce event-level, imbalance-aware evaluation.

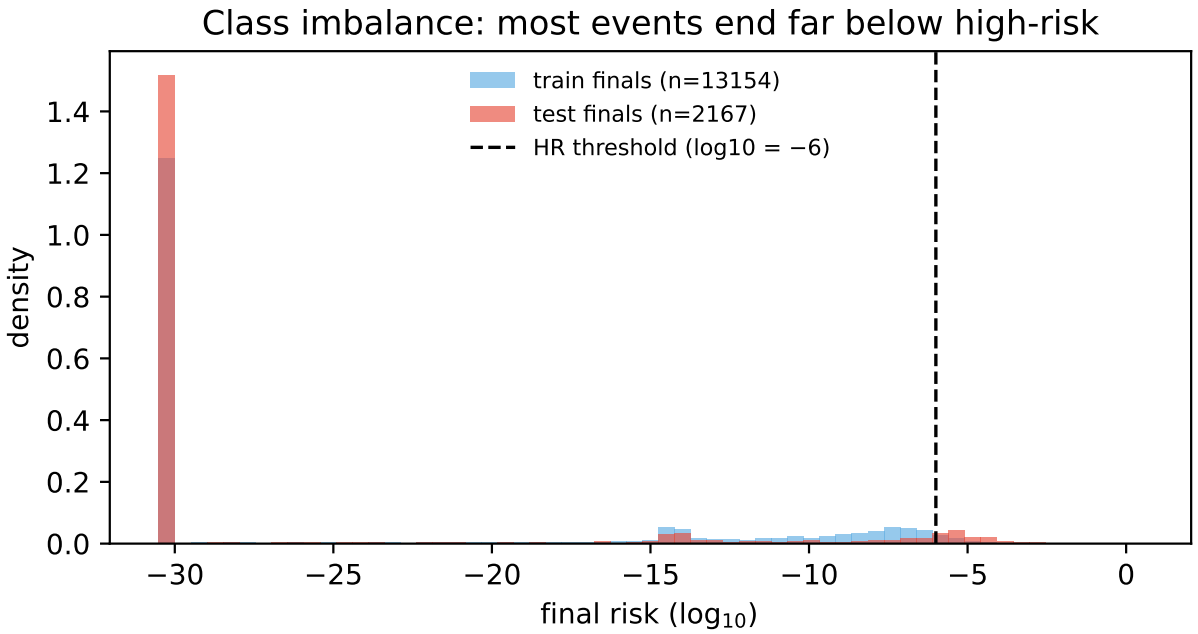


Figure 1: Density of final $\log_{10}$ risk for train-event finals and official test finals. Dashed line: HR threshold $r = -6$. Source: Zenodo DOI `10.5281/zenodo.4463683`.

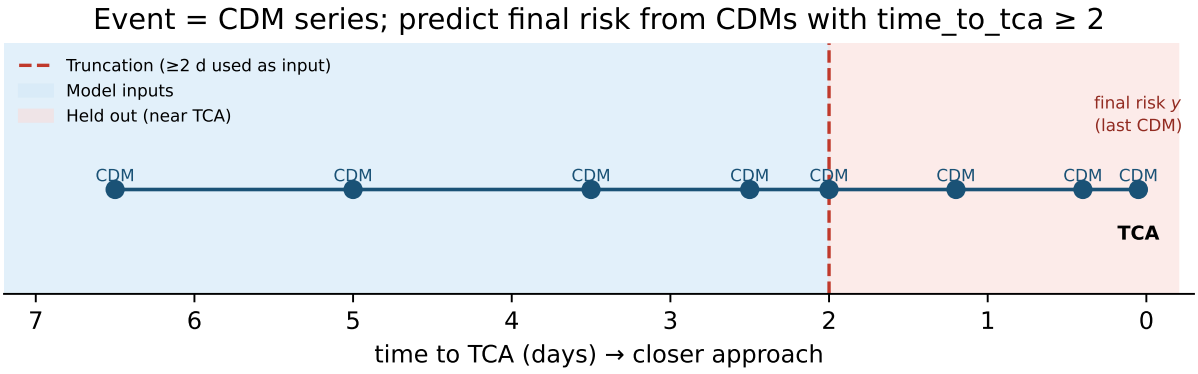


Figure 2: Schematic event: CDM series vs time-to-TCA. Inputs use only messages with time-to-TCA ≥ 2 days; target is final risk near TCA [1, 3].

3 Preliminary Methodology and Results

Evaluation. Scores use official Zenodo test events ($n = 2167$). Inputs: public CDMs with time-to-TCA ≥ 2. Metrics: F_2 (HR vs non-HR), MSE_{HR} on true HR with risk clipped at -6.001, and $L = \text{MSE}_{\text{HR}}/F_2$ [3]. The test set is used here only to reproduce published baselines, not to select models.

Baselines. CRP predicts $\hat{r} = -5$ always. LRP copies the latest eligible early risk (minimum time-to-TCA among CDMs with time-to-TCA ≥ 2). Table 1 and Figure 3: CRP is weak ($L \approx 2.50$); LRP is strong ($L \approx 0.694$, $F_2 \approx 0.739$, $\text{MSE}_{\text{HR}} \approx 0.513$), matching Uriot et al. within rounding [3]. Later models must beat or explain LRP under rare-event metrics.

Table 1: Preliminary baselines on the official Zenodo test set ($n = 2167$; 150 HR). F_2 higher is better; MSE_{HR} and L lower are better.

Model	Split	F_2	MSE_{HR}	L
CRP ($\hat{r} = -5$)	Official test	0.271	0.679	2.504
LRP (latest early risk)	Official test	0.739	0.513	0.694

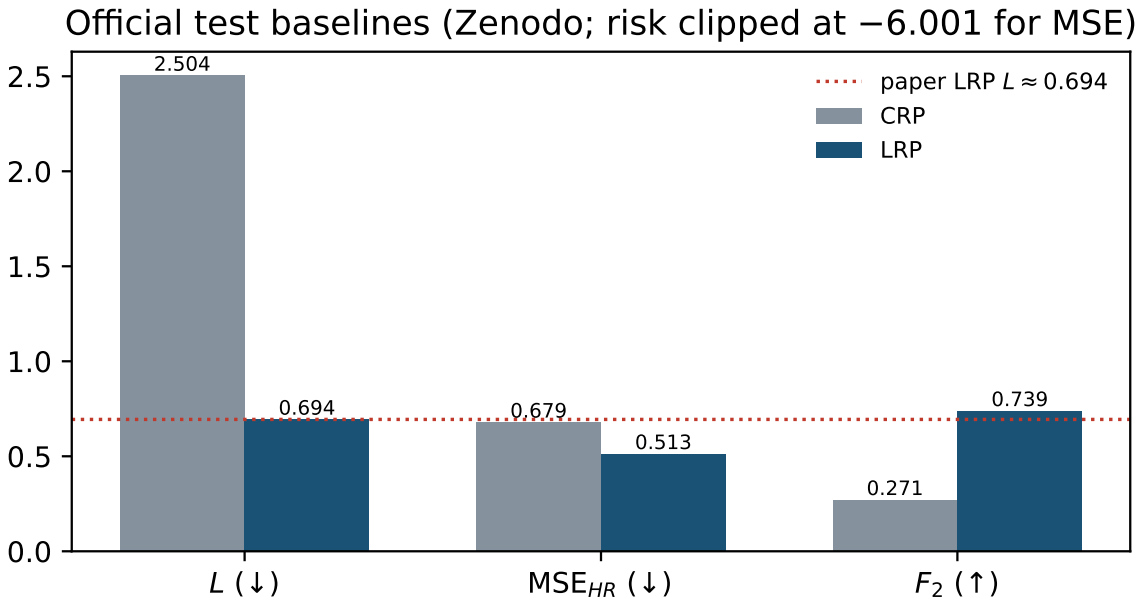


Figure 3: Official-test CRP vs LRP. Dotted line: published LRP loss $L \approx 0.694$ [3].

Planned methods. Much signal sits in the last early risk; residual value is in LRP failures (including jumps) and calibrated probabilities. Path: course models (linear/logistic, tree ensembles, later PyTorch nets) under one event-level protocol, then a two-stage calibrated $P(\text{HR})$ + magnitude head with reliability diagrams and a failure atlas. We will selectively trial out-of-syllabus ideas (temporal encodings, isotonic/temperature scaling, abstention) when they address jump uncertainty—cited as prior art when related to competition winners, never as "re-run Kelvins to win" [3].

4 Next Steps, Reproducibility, and Ownership

Next steps. (1) Freeze metrics; deepen EDA (missions, jump candidates). (2) Features (latest CDM, deltas, aggregates) + course models vs LRP. (3) Two-stage model + calibration + failure atlas; optional ambitious ablations. (4) Fair PyTorch / sequence ablation if justified by jump analysis. (5) Report polish and oral defence.

Risk/contingency. LRP may remain hard to beat and train/test mismatch may destabilise rankings [3]. Contingency: report course models fairly; emphasise calibration and failures even if gains over LRP are small—that outcome still answers the RQs.

Reproducibility. Public GitHub with meaningful history; seeds and environment pinned when stochastic methods appear; Zenodo DOI for data; scripts regenerate tables/figures. Large raw files are not committed.

Declaration of Contributions and AI Use

Teamwork: We jointly chose the ESA early-risk problem, fixed the RQs and evaluation protocol, and reviewed this proposal before submission.

Individual contributions: Pedro Jon Fuentes Garcés: problem framing, research questions, domain narrative, figure captions, and proposal drafting. Mario Fernández Velasco: data handling, CRP/LRP baseline reproduction, metrics, tables, and figure generation from code. Both: protocol decisions and final approval.

AI tools: Cursor (AI assistant) helped with literature organisation, Zenodo download, baseline metric code, figure generation from that code, and a structural draft of this text, which we edited and own. Each member can defend the protocol, CRP/LRP numbers, and two-stage plan. Verification: recomputed official-test CRP/LRP with $L_{\mathrm{LRP}} \approx 0.694$ vs Uriot et al. [3]. AI is not an author.

References

[1] European Space Agency Advanced Concepts Team. Collision avoidance challenge. Kelvins competition platform, 2019. URL https://kelvins.esa.int/collision-avoidance-challenge/. Accessed 11 October 2026.

[2] Thomas Uriot, Dario Izzo, José Martínez-Heras, Francesca Letizia, Jan Siminski, and Klaus Merz. Collision avoidance challenge dataset, 2021. URL https://doi.org/10.5281/zenodo.4463683. Creative Commons Attribution 4.0 International.

[3] Thomas Uriot, Dario Izzo, Luís F. Simões, Rasit Abay, Nils Einecke, Sven Rebhan, José A. Martínez-Heras, Francesca Letizia, Jan Siminski, and Klaus Merz. Spacecraft collision avoidance challenge: Design and results of a machine learning competition. *Astrodynamics*, 5:1–15, 2021. doi: 10.1007/s42064-021-0101-5. Also available as arXiv:2008.03069.